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CFD modeling of CO_2 fixation by microalgae cultivated in a lab scale photobioreactor

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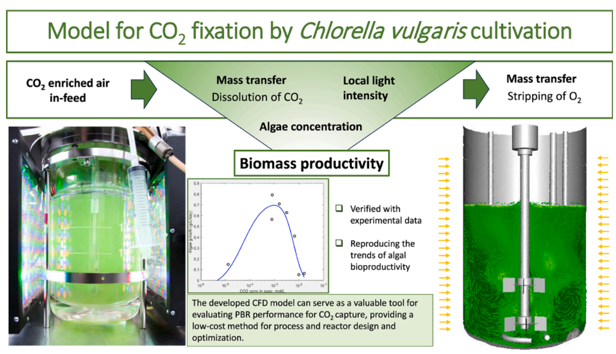
HIGHLIGHTS

- A CFD model for CO_2 fixation by *Chlorella vulgaris* cultivation has been presented.
- A model for algae growth dependency on excess and shortage of dissolved CO_2.
- Verification done with experiments in laboratory scale stirred jacketed reactor.
- The model serves as a valuable tool for evaluating photobioreactor performance.

GRAPHICAL ABSTRACT



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ABSTRACT

The green microalga *Chlorella vulgaris* has been studied for efficient fixation of carbon dioxide, CO_2, from elevated concentrations of in-feed gas flows. In this work, a Computational Fluid Dynamics (CFD) model for algae cultivation including the dependence on CO_2 concentration in liquid has been derived based on cultivation experiments of *Chlorella vulgaris*. The experiments were performed in a laboratory scale cylindrical stirred tank reactor with a range of enriched CO_2 concentrations in-feed (0.04–50 %). The model provided by Béchet et al. (2015) for algae cultivation considering dependence on local light, algae concentration and temperature has been implemented for CFD and modified for comprising CO_2 dependence. The model has been verified with experimental results for the algae productivity showing the proper trends for dissolved CO_2 concentrations and algae concentrations. Thus, the developed CFD model can serve as a tool for evaluating photobioreactor performance for CO_2 capture.

1. Introduction

The application of microalgae for carbon capture has gained growing interest during recent years, not least due to challenges in climate change (Fu et al., 2019; Choi et al., 2020; Madhubalaji et al., 2020; Iglina et al., 2022). CO_2 fixation and storage via microalgae takes place essentially through photosynthesis, which can transform water and CO_2 into organic compounds without additional energy requirements and without secondary pollution. In comparison with terrestrial plants, microalgae are capable of ten-to-fifty-fold higher carbon dioxide fixation (Zhou et al., 2017).

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Nomenclature

a	specific interfacial area (1/m)
α_g	volume fraction of gas
c_{CO2}	CO_2 concentration in suspension (mol/L)
d_p	diameter of gas bubbles
k_L	mass transfer coefficient (m/s)
p_{CO2}	partial pressure of CO_2 in the bubbles
t	time (s)
A, B	empirical coefficients
C_b, C_K, C_I	fitting parameters
$C_{CO2/O2\ L}$	CO_2/O_2 concentration in a liquid phase (kg/m^3)
$C^*_{CO2/O2\ L}$	CO_2/O_2 saturation concentration in liquid (kg/m^3)
$D_{CO2/O2}$	diffusivity coefficient of CO_2/O_2 (m^2/s)
I_{loc}	local light intensity (W/m^2)
I_o	incident light intensity (W/m^2)
He	Henry's constant (Pa L/mol)
K	half-saturation constant at temperature T (W/kg)
L	light path between the position and the reactor external surface (m)
M	molecular weight
$P_m(T)$	maximal specific growth rate at the temperature T (kg (O_2)/ kg-s)
$Pnet$	a net rate oxygen production (kg O_2/s)
PQ	photosynthetic quotient
T	temperature (°C)
$U_{g/l}$	velocity magnitude of gas, g, and liquid, l (m/s)
V	reactor volume (m^3)
X	algal concentration (kg/m^3)
λ	kinetic coefficient of endogenous respiration (kg O_2/kg-s)
σ_X	extinction coefficient (m^2/kg)

Although very promising, CO_2 capture by microalgae still faces limitations. Since undissolved CO_2 is lost by degassing, the challenge is to control the CO_2 amount to be injected (Gonçalves, et al. 2021), i.e., CO_2 must be injected in sufficient, but reasonable quantity for ensuring efficient growth conditions while minimizing loss of CO_2 by degassing. CO_2 supply may be a constraining factor for algae growth if the dissolved CO_2 concentration in the liquid phase is too low, e.g., when air is used as the feeding gas or if the mixing in the photobioreactor (PBR) is insufficient. On the other hand, a high concentration of dissolved carbon dioxide (DCD) can lead to inhibitory culture conditions. Algae's ability to tolerate high CO_2 concentration varies depending on the species. Many microalgae have been found to reach optimal growth under relatively low CO_2 concentration in the range of 2–6 %, as presented by Chen and Xu (2021). With *Chlorella vulgaris*, studies have been conducted also with higher levels (Hulatt and Thomas, 2011), e.g., with as high as 15 % (v/v) (Azhand et al, 2020). However, the value 20 % (v/v) has been reported to be already toxic for Chlorella vulgaris (Li et al, 2023).

Light penetration and distribution inside the PBR are dominant factors affecting the photosynthesis rate, and further, efficient CO_2 fixation during algae cultivations (Huang et al., 2017). Inside a reactor, the light intensity is distributed heterogeneously due to light attenuation, e.g., by absorption and scattering by the algal cells. The effect of gas bubbles has been reported to be of minor importance on the light penetration at algae concentrations greater than 0.5 g/L (Wheaton and Krishnamoorthy, 2012; McHardy et al., 2018). From irradiated surfaces in PBRs light intensity decays sharply with distance due to microalgae self-shading, which is dependent on the algal concentration and cell size. Therefore, algal cells may be exposed to sufficient or even excess lighting near the irradiated surfaces, while in other regions of the reactor, lighting may be insufficient for photosynthetic activity. Mixing plays an important role in ensuring a favorable distribution of the light intensity in the algae cultivation broth by transporting algae cells between the illuminated and dark zones in the reactor, thus providing short light/dark (L/D) cycles. The L/D cycles are widely thought to be beneficial, leading to increased biomass productivities, even with light intensities above the saturation value. However, the influence of the L/D cycle frequency on cell growth is still discussed controversially in literature, especially in industrial scale reactors (Huang et al., 2017; Benner et al., 2022).

The design and scale-up of PBRs for algae cultivations remain a challenge due to immense capital investment and operating cost. Several literature reviews published provide an overview of diverse PBR designs, describing various features and their purpose in microalgae growth and for industrial CO_2 capture, e.g., Pruvost et al. (2016), Razzak et al. (2017), Sadeghizadeh et al. (2017), Fu et al. (2019), Legrand et al. (2021), Sirohi et al. (2022), and Benner et al. (2022). For reducing design costs and improving reactor efficiency, computational fluid dynamics (CFD) serves as an effective tool for modeling and evaluating PBR performance. However, the required models are complex, with relevant phenomena being highly coupled and interacting across widely separated timescales (Gao et al., 2017). So far, CFD modeling has been applied mostly for studies of hydrodynamics, such as mixing, hold-up, and mass transfer of CO_2 from gas phase to liquid. The phenomena of photosynthesis, algae growth, mass transfer of CO_2 from liquid into algae, and mass transfer of oxygen, O_2, produced during photosynthesis have been modeled to a lesser extent (Kommareddy and Ananthula, 2017; Sadeghizadeh et al., 2018). Luzi and McHardy (2022) presented a comprehensive review of the state of the art of describing PBR through CFD.

Most studies presented in literature describing the kinetics of microalgae growth were based on set CO_2 concentrations, so that the rate constant for the photosynthetic reaction is dependent on the algae biomass, but not on the CO_2 concentration (Adamczyk et al., 2016; Sadeghizadeh et al., 2018; Azhand et al, 2020). In addition, the studies have been mostly conducted with air-lift type reactors.

In this work, a CFD model for CO_2 fixation by algae cultivation has been derived and verified based on experiments with *Chlorella vulgaris* with a range of infeed gas CO_2 concentrations from 0.04 % to 50 % in stirred tank, which is a conventional aerated bioreactor. With stirred reactor, the mixing is very effective so that the medium is homogenous for model development purposes. The model provided by Béchet et al. (2015) for algae growth with the dependency on light intensity, algae concentration, and temperature has been implemented for CFD, and further modified for comprising bioproductivity rate dependency on CO_2, i.e., the effect of dissolved CO_2 concentration on algae growth from deficiency to excess. The developed model includes mass transfer of carbon dioxide and oxygen between gas and liquid phases and allows evaluation of the PBR's performance for CO_2 capture.

2. CO_2 fixation experiments in laboratory scale

Algal cultures of *Chlorella vulgaris* (CCAP211/11B, obtained from the Culture Collection of Algae and Protozoa (CCAP) Oban, Scotland, United Kingdom) were grown in a 2 L jacketed glass reactor (2 L Biostat B plus Bioreactors, Sartorius Stedim, Goettingen, Germany) with double stirrer. The reactor geometry is shown in Fig. 1. The reactor's inner diameter is 133 mm and total reactor height 242 mm, with a curved bottom. The distance between the reactor bottom and lower stirrer disc is 28.4 mm, distance between the stirrer discs 37.1 mm and diameter of both stirrers is 52.5 mm consisting of six blades. The shift between the position of the stirrers is 8.3 degrees for the upper stirrer. Dual Rushton impeller configuration was chosen, because its two double loops flow

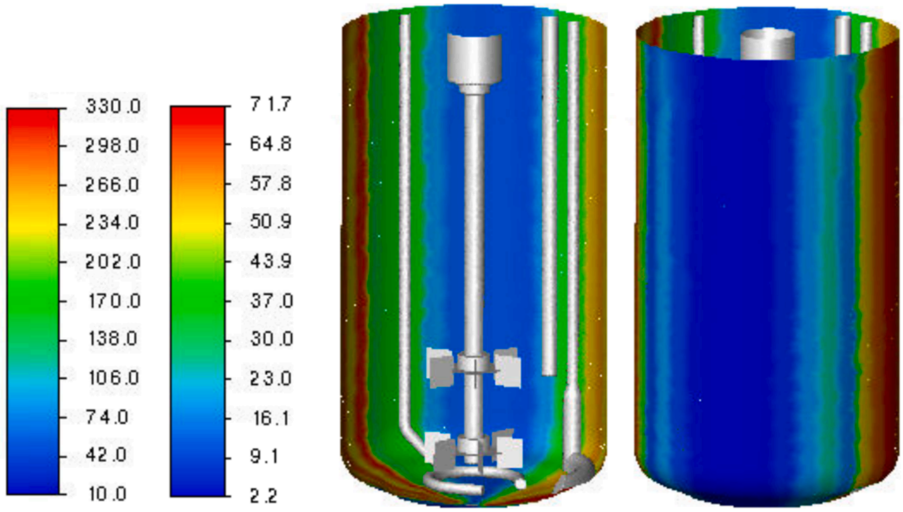

Fig. 1. Reactor geometry and measured local light intensity on the inner reactor surface; scales from the left: μmol/m^2-s, W/m^2.

pattern extends to the whole vessel volume, providing very effective mixing without stagnant areas, thus enhancing the mass transfer between the phases.

The height of the suspension was 160 mm from the middle of the reactor bottom. The reactor set up consisted also of probes for measuring pH and temperature, as well as a sampling pipe. The reactor was operated with a rotational speed of 100 rpm and a constant temperature of 24°C. pH was maintained at 7 through addition of 1 M NaOH and 15 % H_3PO_4, respectively. From algae cultivation point of view, 100 rpm is a compromise between good mixing and not introducing shear stress on the algae cells. Chlorella vulgaris was cultivated in Modified Acid Medium (MAM, Olaueson and Stokes, 1989). The cultivations were initiated by removing approximately 90 % of the entire culture from the previous experiment and diluting with fresh medium to obtain initial optical density of 0.15 – 0.30 (600 nm).

For efficient algal growth, an optimal amount of light is important. Two flat LED panels (20 × 20 each; SL 3500 from Photon Systems Instruments; Drasov, Czech Republic) at opposite sides of the reactor provided continuous lighting. The LEDs produced a mixture of blue (440–460 nm), green (520–540 nm), and red (620–645 nm) light. During the experiments of 175 h, the light panels were set to 20 % intensity. The light intensity was measured with a quantum sensor connected to a radiometer (Li-Cor. Inc., USA) at the middle height of the reactor inner surface and modeled by drawing the light intensity in vertical direction, as shown in Fig. 1. A range of 60–200 μmol/m^2/s, that is 13–44 W/m^2, has been reported as an optimal light intensity for growth of *Chlorella vulgaris* (Wang et al., 2014). The light maintenance point (i.e., the irradiance below which photolimitation occurs) for *Chlorella vulgaris* is around 1–2 W/m^2, and the light saturation intensity (i.e., the irradiance above which photoinhibition occurs) is about 54 W/m^2 (Degen et al., 2001).

The cultures were supplied with gas feeds ranging from air (CO_2 concentration 0.04 %), and further with six enriched CO_2 concentrations, namely 2.5 %, 5 %, 10 %, 20 %, and 50 % (v/v). The gas was fed into the bottom of the reactor through a sparger with 14 holes of 0.6 mm diameter, see Fig. 1. The flow rate was controlled through a mass flow controller to ensure that gas flow remained constant at 0.4 L/min (NTP) and gas mixture at the set level throughout the experiment. Concentrations of gas components were measured from the input and output by mass spectrometry (Prima Pro Process mass spectrometer, Thermo Scientific, UK). The average bubble size was analysed from images of the setup to be 0.52 mm.

The measured optical density at 600 nm (OD600 values, Shimadzu UV – 1201 Spectrophotometer, AMERICA North) is presented in Fig. 2.

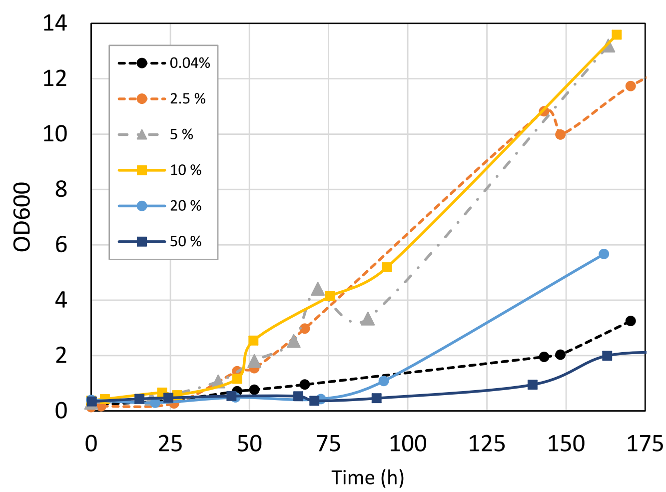

Fig. 2. Measured OD600 values from the algae cultivation as a function of time with varying CO_2 concentration of in-feed gas.

The ratio of OD600 to algal dry mass was determined experimentally to be 0.185. The average inlet gas concentrations for nitrogen, oxygen, and carbon dioxide in percentages are presented in Table 1. Further, the average changes in concentrations for oxygen and carbon dioxide between inflow and outflow values are presented in relation with nitrogen, which is considered as an inertial gas component.

According to the experimental results, the optimal CO_2 concentration level for cultivation of *Chlorella vulgaris* was between 2.5 and 10 % (v/v), see Fig. 2. With higher levels of CO_2 concentration, the

Table 1
Experimental values for gas in-feed, and the result for the change in oxygen and carbon dioxide concentrations between inflow and outflow, in average.

	In-feed gas			Outflow		
Case	Nitrogen	Oxygen	Carbon Dioxide	Nitrogen	Oxygen	Carbon Dioxide
0.04 %	78.2 %	20.8 %	0.047 %	100.0 %	100.1 %	63.9 %
2.5 %	78.4 %	18.2 %	2.6 %	100.0 %	100.5 %	97.0 %
5 %	78.6 %	15.7 %	4.9 %	100.0 %	100.9 %	97.7 %
10 %	79.0 %	10.7 %	9.8 %	100.0 %	101.2 %	98.8 %
20 %	62.4 %	16.6 %	20.2 %	100.0 %	100.6 %	99.4 %
50 %	40.0 %	10.6 %	49.0 %	100.0 %	100.1 %	99.9 %

bioproductivity of the algae was clearly decreased. Moreover, bioproductivity was slow with lower values of CO_2 concentration. This is in accordance with the findings in literature, e.g. Li et al. (2023).

3. Microalgae productivity model

3.1. Model applied for biomass productivity

The biomass productivity model developed by Béchet et al. (2015) includes the phenomena of endogenous respiration as well as light and temperature acclimation. The model by Béchet et al. (2015) formulates a net oxygen production rate expressed as difference between the rate of chemical energy harvested and the rate of respiration (kg O_2/s). In Eq. (1), the local oxygen productivity by Béchet model is expressed in a form applicable with CFD:

$$\frac{dP_{net}}{dV} = P_m(T)\frac{\sigma_X I_{loc}}{K(T) + \sigma_X I_{loc}}X - \lambda X \tag{1}$$

where I_{loc} is the local light intensity (W/m^2), T is the temperature (°C), V is the reactor volume (m^3), K (W/kg) represents the half-saturation constant at temperature T, σ_X is the extinction coefficient (m^2/kg), and X is the algal concentration (kg/m^3). $P_m(T)$ denotes the maximal specific growth rate at the temperature T (kg(O_2)/kg-s) and λ (kg(O_2)/kg-s) the kinetic coefficient of endogenous respiration. The model is parametrized for *Chlorella vulgaris* (Béchet et al., 2015).

For expressing P_m (kg(O_2)/kg-s) as biomass productivity P'_m (kg/kg-s), a conversion coefficient of 0.76 was applied (Béchet et al., 2015). Similarly, the value of λ (kg(O_2)/kg-s) was presented as λ' (kg/kg-s) in light conditions with conversion coefficient 0.76, and in dark conditions with coefficient 0.94.

The carbon capture rate is computed from the oxygen net rate production, Eq. (1), by photosynthetic quotient PQ, which is defined as the ratio of the oxygen net productivity n(O_2) (mol/s) over the net carbon capture rate n(C) (mol/s). In here, PQ = 1.44 according to Becker (2007) has been applied.

3.2. Local light intensity in PBR

The productivity model in Eq. (1) assumes that a single algal cell harvests energy only as a function of the light intensity the algae cell is exposed to, and is based on the Monod model that has been applied for describing algal productivity under a broad range of operational conditions and designs. The approach is considering the response of microalgae to light intensity and the distribution of the light intensity as a function of the depth and biomass concentration within the culture, but it is not capable to capture the photoinhibition due to the excess light. A non-monotonic expression, such as a Haldane function, would consider photo-inhibition due to excess light intensity (Krichen et al., 2021); however, the microalga *Chlorella vulgaris* is quite tolerant regarding lighting, as shown by Degen et al. (2001) and Madhubalaji et al. (2020). Degen et al. suggested that photoinhibition can be avoided by keeping the irradiance at about 250 μmol/m^2/s in dilute cultures. Madhubalaji et al. applied 400 μmol/m^2/s at higher concentrations of over 0.2 g/L with good response to the bioproductivity.

For describing the local light intensity, a modified Beer-Lambert law, Eq (2), has been applied in the model of Béchet et al. (2015):

$$I_{loc}(I) = I_0 \exp(-\sigma_x X l). \tag{2}$$

The extinction coefficient σ_x, is a measure of the rate of transmitted light via absorption for a medium, and is defined by Béchet et al. (2015) as

$$\sigma_x = AX^B \tag{3}$$

where A and B are empirical coefficients (A = 117.4 + -1.6 m^2/kg; B = −0.200 + -0.06). l denotes the light path between the local position and the reactor external surface (m), and I_o the incident light intensity (W/m^2).

The Beer-Lambert model does not include light scattering. Wágner et al. (2018) investigated two models for light distribution in algae cultures: Schuster's law including light scattering and Beer-Lambert expression without light scattering. They concluded that the prediction by Beer-Lambert equation was sufficient in reactors intended to be used at high biomass concentrations, as is typically the case in PBRs, e.g., in a case with total solids content higher than 0.04 g/L, and reactor diameter 140 mm or wider.

3.3. Modifying the bioproductivity model for CO_2 dependency

The CFD model presented here for the fixation of CO_2 by *Chlorella vulgaris* is based on the model developed by Béchet et al. (2015), Eq. (1). In our work, the model has been enhanced to also include algae growth rate dependency on CO_2 concentration in the culture broth; limitation of growth rate due to shortage or excess of dissolved CO_2 available.

For expressing the biomass productivity limitation by CO_2 levels, we suggest the following model $F(c_{CO2})$ based on relationships presented by Nauha and Alopaeus (2013), and Zhang et al. (2021)

$$F(c_{CO2}) = C_t\left(\frac{C_{CO2}}{C_K + c_{CO2}}\right)\left(\frac{C_I}{C_I + c_{CO2}^2}\right) \tag{4}$$

where c_{CO2} is carbon dioxide concentration in the culture (mol/L). With low carbon dioxide levels, the algal growth rate is limited by the first part of the model. With high carbon dioxide levels, inhibition occurs according to the second part of the model. The model is obtained by fitting the experimental maximum specific growth rate as a function of CO_2 concentration in the liquid through adjusting parameters C_K and C_I. These results are then adjusted with parameter C_t to the model for algae cultivation proposed by Béchet et al. (2015), which was derived based on experiments with 2 % (v/v) CO_2 inlet gas concentration. In our work, the respiration rate was assumed to be independent of the CO_2 concentration in the algae culture, as in respiration, the cells consume oxygen in the presence of light and release carbon dioxide instead of fixing.

In this work, each algal cultivation lasted more than 150 h, so that the broth could be considered saturated in respect of CO_2 concentration of the gas bubbles. Therefore, for the purpose of model parameter fitting, the suspension concentration values c_{CO2} are derived from the outflow gas CO_2 concentration. In addition, the difference between the inflow and outflow values was small and stabilized.

The modified algae productivity model incorporates the dependency of algal growth on CO_2 concentration (mol/L) in the liquid, c_{CO2}:

$$\frac{dP'_{net}}{dV} = C_t\left(P'_m(T)\frac{\sigma_X I_{loc}}{K(T) + \sigma_X I_{loc}}X\left(\frac{C_{CO2}}{C_K + C_{CO2}}\right)\left(\frac{C_I}{C_I + {C_{CO2}}^2}\right)\right) - \lambda X \tag{5}$$

The obtained parameter values are C_t = 1.175, C_K = 5.2e-5 (mol/L) and C_I = 3.5e-5 (mol^2/L^2). The relationship between algal growth and CO_2 concentration in the algae culture, c_{CO2}, as described by the modified model, is presented in Fig. 3. The experimental values are denoted by open circles.

3.4. Modeling mass transfer of CO_2 and O_2 between phases

Proper mass transfer between the gas and liquid phases is a necessity, as microalgae can only make use of dissolved CO_2 as a carbon source. In mass transfer limited processes, the CO_2 dissolution rate limits microalgae growth. Large interface areas between the gas and culture medium ensure good mass transfer. Thus, bubble size is a crucial parameter to improve the performance of gas–liquid reactors, particularly in situations where reactions are mass transfer limited (Ding et al., 2016; Hanotu et al., 2017). The small bubbles provide a large surface area for mass transfer between the phases. For providing a homogenous bubble flow system, the sparger design is essential (Hanotu et al., 2017).

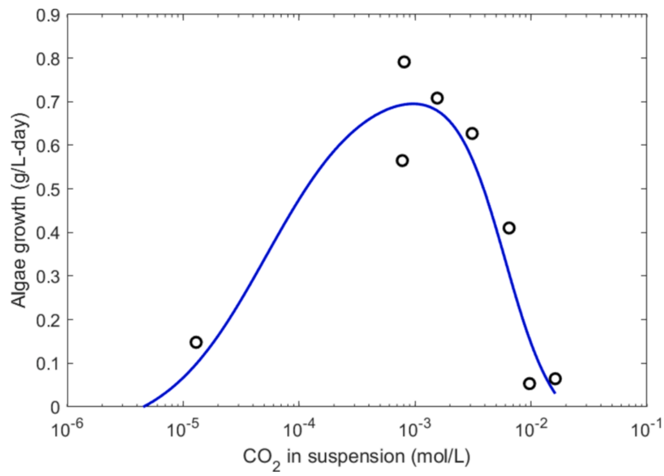

Fig. 3. Algae productivity rate (g/L-day) computed with Eq. (5) as a function of CO_2 concentration in algae culture. Experimental values are denoted by open circles.

The residence time of small bubbles is longer in the suspension, as they follow the fluid flow due to negligible buoyancy, thus strengthening mass transfer (Ding et al., 2016). In a stirred reactor, the impellers are keeping the suspension in motion; Bubbles are following this motion pattern, therefore increasing the residence time of also somewhat bigger bubbles in a suspension. Further, mass transfer between gas and suspension is dependent on the concentration gradient, thus, mixing is an effective method to enhance interphase mass transfer. Ding et al. (2016), and Barahoei et al. (2020), proposed that small bubbles with high CO_2 fractions (> 5 % v/v) are more favorable for microalgal growth due to their high retention time in liquid and dissolution efficiency.

The gas–liquid mass transfer (kg/m^3-s) is modeled through

$$\frac{dC_{CO2_L}}{dt} = (k_L a)_{CO2}\left(C^*_{CO2_L} - C_{CO2_L}\right) \quad (6)$$

where C_{CO2L} is the CO_2 concentration (kg/m^3) in the liquid phase and C^*_{CO2L} is the CO_2 saturation concentration (kg/m^3) in liquid given by Henry's law

$$C^*_{CO2_L} = M_{CO2}\frac{p_{CO2}}{He} \quad (7)$$

where p_{CO2} is the partial pressure of CO_2 in the bubbles, M_{CO2} its molecular weight and He Henry's constant, which depends on the temperature and salinity of the culture medium. The effect of hydrostatic pressure was considered in solubility by the local partial pressure.

The dissolution rate of CO_2 from bubble to microalgae suspension is determined by the CO_2 mass transfer coefficient k_L, that is modeled by Higbie's penetration theory (Higbie, 1935):

$$k_L = \frac{2}{\sqrt{\pi}}\sqrt{\frac{D_{CO2}|U_g - U_l|}{d_p}} \quad (8)$$

where D_{CO2} is the diffusivity of CO_2, and $U_{g/l}$ the velocity of gas g, and liquid l. d_p is the diameter of the gas bubbles.

The specific interfacial area a, between continuous and dispersed phases is given as

$$a = \alpha_g\frac{6.0}{d_p} \quad (9)$$

where α_g is the holdup or the volume fraction of the dispersed gas phase.

Eq. (8) implies that the mass transfer coefficient is directly related to the bubble rising velocity and inversely related to the bubble diameter. Therefore, as the interfacial area between the phases, Eq. (9), is also inversely related to the bubble diameter, the mass transfer is strongly dependent on the bubble diameter; the smaller the bubble, the better the mass transfer rate. Coalescence of the bubbles produces larger bubbles with decreased interfacial surface, thus reducing the mass transfer between the continuous and dispersed phases. However, the rising velocity of the larger particle is higher, also affecting the dissolution rate.

The presence of salts in the liquid phase, i.e. dissolved ions e.g. due to pH control, decreases CO_2 solubility. For computing the mass transfer of CO_2 in a solution corresponding to the *Chlorella vulgaris* culture containing 2.39 g of salt/kg of water, the applied values are $D_{CO2} = 1.99 \times 10^{-9}$ m^2/s, and $He_{CO2} = 30.9 \times 10^5$ Pa L/mol at 25 °C; similarly, oxygen mass transfer between the phases was computed with the values $D_{O2} = 2.41 \times 10^{-9}$ m^2/s, and $He_{O2} = 952 \times 10^5$ Pa L/mol (Ndiaye et al., 2018; Roustan, 2003; Schumpe, 1993).

Efficient mass transfer is important also for removing dissolved oxygen DO, released as a byproduct of photosynthesis, from the broth, as high levels of DO may be harmful for the microalgae. Small bubbles with large interface areas are efficient in removing oxygen from the culture. However, according to Kazbar et al. (2019), the DO limit for loss of biomass productivity of *Chlorella vulgaris* microalgae is as high as 30 g/m^3.

4. Computational methods and boundary conditions

The CFD computations were carried out with ANSYS Fluent software release 2022 R2 with Linux cluster using 36–64 cores. The size of the computational mesh was 2,69,600 cells. Polygonal grid was used and preferred over tetrahedral grid for lower number of cells and less numerical diffusion to include the details of the modeled geometry correctly. Structured mesh layer in wall-normal direction was used on the walls with averaged cell wall distance of 0.419 mm. The grid was optimized by improving its skewness and refining it in the vicinity of the relevant parts of the modeled geometry such as the impellers, probes, and gas in-feed pipe. The maximum skewness value of the grid was 0.62 and its average orthogonal quality value was 0.97. A quite dense mesh was applied due to the details of the geometry and the sliding mesh technique applied for rotating stirrers.

The algae culture was described by Eulerian two-phase model, with the aqueous primary phase and the secondary phase comprising the gas bubbles. The flow was considered as turbulent, with Re approximately 48000, but not fully turbulent. Therefore, a Shear-Stress Transport (SST) k-ω model was applied for modelling turbulence, as this approach is capable of capturing also the small Reynolds number flows and is generally thought to perform better near walls. Species transport equations were solved for both phases, which contained chemical species of carbon dioxide and oxygen. The dispersed gas phase additionally included nitrogen. The algae were described as a specie in aqueous phase.

Second order space discretization was used in the solution of the convective terms in the momentum and dispersed phase volume fraction equations, and first order space discretization in turbulence and species transport equations to ensure convergence. The pressure–velocity coupling was solved with phase coupled SIMPLE algorithm, with a first-order time discretization scheme.

Of the interfacial forces resulting from the interaction between continuous and dispersed phases, drag force was considered as the dominant force. In here, the dispersed phase, i.e., gas bubbles in the suspension region of the reactor became the continuous phase in the free gas domain above the broth. Therefore, for the drag force, the symmetric model of ANSYS Fluent was applied, with the coefficient C_D by Schiller and Naumann (1933). Turbulent dispersion accounting for the interphase turbulent momentum transfer was considered through expression developed by Simonin and Viollet (1990). Other interfacial non-forces, such as lift or wall lubrication forces, were neglected as insignificant compared to the drag force in the case of the stirred tank (Scargiali et al., 2007).

The models for algae cultivation as described in Sections 3 and 4 were implemented with User Defined Functions (UDF) in ANSYS Fluent. The physical material parameters applied for the algae cultivation process were set to 1025 kg/m^3 (Coons et al. 2014, Kommareddy and Ananthula, 2017) for suspension density and 9.1e-4 kg/m-s (Kestin et al., 1978) for viscosity. The parameters for algae growth rate in Eq. (5), P_m = 5.496e-6 kg(O_2)/kg-s, K = 3909 W/kg, and λ = 1.998e-6 kg (O_2)/kg-s were interpolated at temperature 24 °C from the values presented by Béchet et al. (2015). Stirring in our experiments occurred at a rotating speed of 100 rpm clockwise.

For the bubbles, a constant average diameter d_p = 0.52 mm based on the experiments was adopted in favor of reasonable computation time, although the application of population balance computation for bubble size distribution would have been a more sophisticated method. The use of constant diameter is justified in this case with double stirrer, as small bubbles occupy most of the reactor domain.

5. CFD modeling results and discussion

Several cases with elevated CO_2 concentrations, corresponding to the experimental setups, were chosen for CFD modeling. Algae cultivations are time dependent processes that last several days. Thus, for reduction of computational effort required, selected time points were chosen for modeling. The inlet gas concentration and algae concentration were selected corresponding to an experimental measurement time instant and provided boundary conditions for the CFD modeling. Initialization of CO_2 and O_2 molar concentrations in suspension assumed saturation regarding the outflow value according to the corresponding experimental case. In Table 2, the process parameters for the studied cases are presented.

The experimental measurements show variation between the measurement points, as seen in Fig. 2 and Fig. 4a. For obtaining a function for experimental measurements of bioproductivity rate, a curve was fitted with the concentration values at measured time instants (presented with solid line in Fig. 4a), and derivated in time. The productivity rates at sample time instants are then computed and presented for comparison with modeled results in Fig. 4b with filled squares. CFD results for algal productivity rate (g/L-day) computed with Eq. (5) are presented in Fig. 4b with open squares. In addition, the results presented in Adamczyk et al. (2016) for cultivation with *Chlorella vulgaris* with CO_2 concentration of 4 % (v/v) and 8 % (v/v) are shown for comparison in Fig. 4a. The results from literature are converted to obtain the microalgae growth rate shown in Fig. 4b. The algae growth rates obtained both in our experiments and from literature are comparable.

The cultivation medium used by Béchet et al. (2015) and Adamczyk et al. (2016) was BG-11 and BBM, respectively, and in their work the mixing was introduced by air feed. In our work the medium was MAM. It is assumed that other nutrients in the medium are not limiting microalgae growth, and thus the growth is considered as dependent on the CO_2 concentration of medium. As seen in Fig. 4a and b, the growth is comparable despite differences in media or mixing methods, especially at higher biomass concentrations. In the beginning of experiments, when biomass is still low, the differences are larger, which can be explained by differences in light penetration and circulation. In the model development, only the behavior of growth rate dependency on the CO_2 was considered.

Table 2
Selected cases for CFD modeling.

CO_2 concentration in gas	Algae concentration, g/L	Gas density, NTP, kg/m3	Gas mass flow rate, g/s
2.5 %	0.094	1.293	0.00862
2.5 %	0.55	1.293	0.00862
5 %	0.2	1.305	0.00870
20 %	0.2	1.416	0.00944

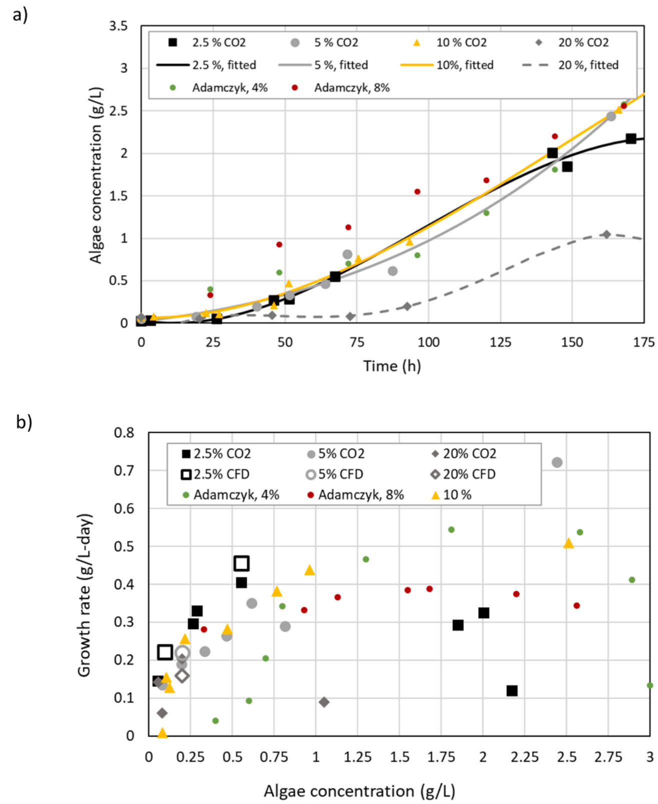

Fig. 4. A) Fitted curve (solid line) on the experimental results (marks). b) CFD results (open marks) compared with experimental results (filled marks) for algae productivity rate (g/L-day) as a function of algae concentration computed by eq.(5). The results by Adamczyk et al. (2016) are shown for comparison.

Fig. 4b shows the agreement between CFD model including the dependency on dissolved CO_2 concentration with the experimental results for the microalgae growth rate. In overall, the trends of the solution CO_2 concentration and algae concentration are reproduced, although the model is slightly overestimating the bioproductivity. The excess illumination not considered by the monod type model for growth, or the scattering not captured by the Beer-Lambert model for light attenuation might affect the results. In our experimental set-up, the local irradiation was at maximum 330 μmol/m^2/s, which was lower than 400 μmol/m^2/s applied by Madhubalaji et al., 2020. Nevertheless, the developed CFD model can consider the local variating environment of algae cultivation in reasonable way for comparison of, e.g., different process geometries for CO_2 fixation by modeling methods.

Examples of computed instantaneous results for the PBR are presented in Figs. 5 and 6 for the case with 2.5 % CO_2 inlet gas concentration and 0.55 g/L algae concentration. Fig. 5a, left side, shows the volume fraction of the gas phase inside the suspension up to 5 %. The average volume fraction was around 0.6 % in the modeled cases. The distribution of the mass transfer rate coefficient, k_La, shown in Fig. 5a, on the right is following the gas volume fraction distribution shown on the left.

The bubbles fed by the sparger enter the culture broth above the sparger limiting the region occupied by bubbles. The bubbles were not entering to the bottom of the reactor, and thus, the mass transfer between the phases was limited to the upper suspension. In our case, the dissolved CO_2 was distributed by mixing with stirrers. Inside the cultivation broth, stirring kept the bubbles in the flow, but at the top of the suspension, bubbles were separating from the fluid. The velocity difference between the phases was on average 0.06 m/s for all the modeled cases, as the bubble size and the rotation speed were constant. Also, the gas volume flow rate was kept constant over all studied cases. Therefore,

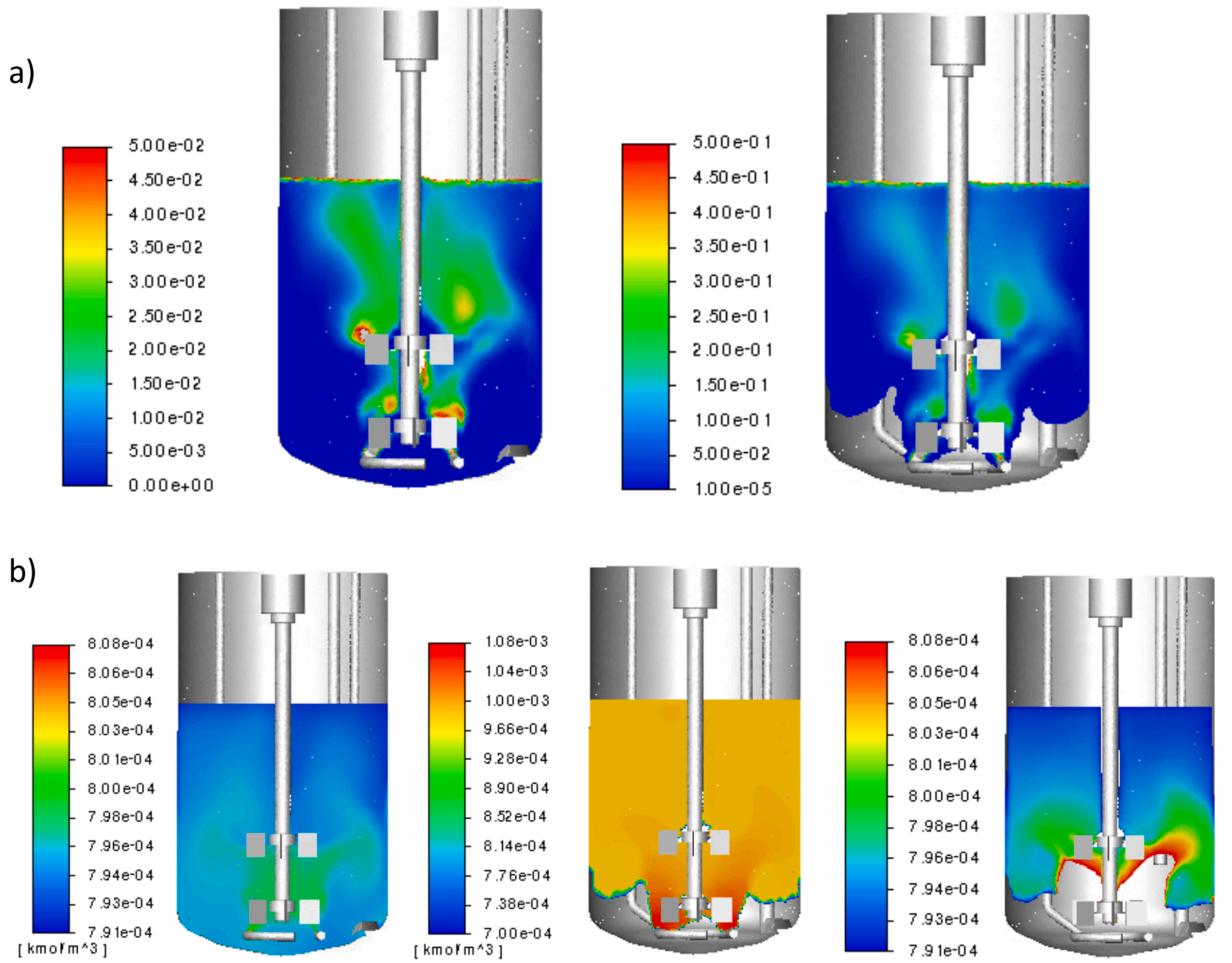

Fig. 5. A) Left: Distribution of volume fraction of gas phase; Right: Mass transfer rate coefficient k_La of CO_2 (1/s). b) Distribution of CO_2 molar concentration (kmol/m^3). Left: dissolved CO_2 molar concentration in suspension; Middle: CO_2 molar concentration in gas phase; Right: saturation concentration of dissolved CO_2 in the suspension.

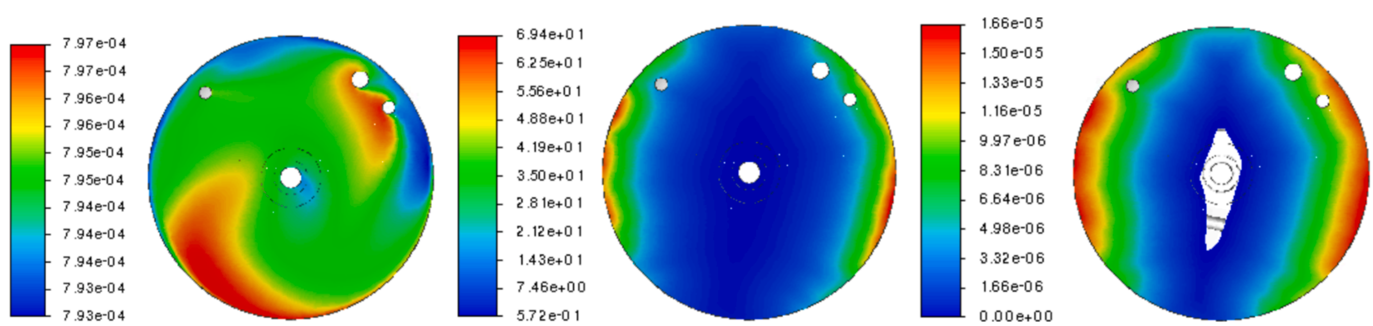

Fig. 6. Left: dissolved CO_2 concentration distribution (kmol/m^3); Middle: distribution of light intensity (W/m^2); Right: computed algae production rate (kg/m^3-s).

the resulting average mass transfer rate coefficient $k_La = 0.04$ 1/s was similar in all the cases.

The gas–liquid mass transfer rate is the product of two terms: the mass transfer rate coefficient k_La, and the driving force $\left(C^*_{CO2_L} - C_{CO2_L}\right)$ (see Eq. (6)). With high concentration gradient and high interfacial area, mass transfer is fast, and effectively reduces the CO_2 concentration in bubbles as CO_2 is relatively highly soluble in water. In Fig. 5b on the left, the modeled results are presented for the CO_2 molar concentration in suspension (on the left) and the saturation concentration (on the right) computed from local bubble CO_2 concentration (at the middle) with Henry's law.

As shown in Fig. 5b, around the gas feeding region, the suspension CO_2 molar concentration is lower than the saturation value, resulting in fast mass transfer occurring from bubbles to fluid. As the bubbles rise in a reactor releasing CO_2, the saturation concentration decreases. CO_2 molar concentration in the suspension and the saturation concentration are close to each other in the top region of the broth implying the reduced mass transfer rate due to a reduction of the driving force. Therefore, in this region, it is also possible for CO_2 mass transfer to occur from suspension back to the bubbles. Through efficient mixing, from regions of fast mass transfer the dissolved CO_2 is spread to the lower mass transfer regions, giving rise to the possibility of CO_2 suspension concentrations higher than the saturation value.

It is worth noting that the concentration of dissolved CO_2 in the suspension is determined by the gas bubbles' CO_2 concentration. If the bubbles are depleted of CO_2, it is not possible to elevate the dissolved CO_2 concentration in the suspension. Therefore, the dimension of the reactor needs to be optimized in regard to the dissolution time of the bubble CO_2-concentration. In here, the bubble dynamics need to be considered as well: as the bubbles coalescence, the interfacial surface area decreases, thus reducing also the mass transfer rate between the phases. However, the increased rising velocity of the larger bubbles strenghtens the mass transfer rate. In our case, we did not consider coalescence of the bubbles. In the upper parts of the reactor, where back-transference of dissolved CO_2 takes place, the mass transfer coefficient would have been smaller if bubble coalescence producing larger bubbles was considered.

Fig. 6 shows the CFD results at the horizontal cross section of the reactor cylinder at the height of 10 cm from the bottom, right above the upper propellers. The instantaneous results are computed for the 2.5 % CO_2 inlet gas concentration case. The distribution of dissolved CO_2 is fairly homogeneous, due to the good mixing of the suspension. Behind the probes (white dots close to the perimeter of the reactor), the mixing is not as thorough, resulting in lower CO_2 concentrations. The differences in local CO_2 concentrations are so small, however, that the effect on local algae production rates is not noticeable. The production rate distribution is correlated tightly with the distribution of light intensity (see Fig. 6, middle). In the center of the reactor, the light intensity is limiting photosynthesis and therefore algae growth, and respiration occurs instead, shown by a white region on the right, Fig. 6. In this region, the light intensity is less than 1–2 W/m^2, which was the light maintenance point for *Chlorella vulgaris*.

With CFD modeling, mass transfer of oxygen was also followed between the phases. The modeled molar concentration of DO, produced by photosynthesis, was highest at the suspension surface, around 6 to 7 g/m^3 in maximum, thus clearly under the reported toxic value for *Chlorella vulgaris*.

For evaluating CO_2 fixation efficiency, the CO_2 concentration in the gas outlet was monitored during the CFD modeling. With the CO_2 enriched cases, more than 90 % of the fed CO_2 ended up exiting the reactor, which corresponds to the experimentally determined values presented in Table 1. With the process not limited by mass transfer between the phases, the suspension is already saturated in relation to the CO_2 concentration of the bubbles, and therefore CO_2 fixation efficiency cannot be improved by increasing gas–liquid mass transfer. Instead, CO_2 fixation is limited by the algae growth rate. The fixation efficiency is directly proportional to the bioproductivity, and thus depends, e.g., on the algae biomass concentration as well as light intensity available to the cells. The faster the algae grow, the higher is the CO_2 fixation rate. However, enriched CO_2 concentration in the gas bubbles is a prerequisite for elevated concentration of CO_2 in the culture broth.

CFD gives a valuable insight into the cultivation process with low cost. With the robust model provided in this paper, the performance of different geometries for PBR can be compared for optimizing the CO_2 biofixation. The results show the local distribution of key factors affecting microalgal growth, e.g. distribution of dissolved CO_2, light intensity, and algae concentration, and thus, the local algae bioproductivity rate. Therefore, e.g. reactor geometry regions of shortage of light or dissolved CO_2 can be estimated with the model. To our knowledge, no such algae growth model with dependency of dissolved CO_2 has been presented, that also includes inhibition of bioproductivity due to excess CO_2. Inclusion of this phenomenon is important for finding the best conditions for CO_2 biofixation. With the developed CFD model presented here, the efficiency of the process mixing in regard to dissolved CO_2 concentration distribution can be exposed, independent weather the mixing is, e.g., due to the air bubbles in the air-lift reactors, or the agitator. Mixing is an important factor in cultivation processes, thus the developed CFD model can serve a tool for PBR design for CO_2 biofixation by microalgae. In addition, the effect of bubble size on mixing as well as the dissolution process can be studied. In this work, bubble coalescence was not included but is a phenomenon to be considered especially in the case where the solution mixing is provided by gas bubbles only. Further, the environment of the experimental setup was kept constant between the cultivations that created the data for model development.

6. Conclusions

This paper presents a CFD model for *Chlorella vulgaris* cultivation describing gas–liquid mass transfer of carbon dioxide and oxygen, and algae growth in varying culture conditions with dependency on light intensity, algae concentration, and temperature, and comprising the dependency on dissolved CO_2. Experiments were conducted for *Chlorella vulgaris* with varying in-feed gas CO_2 concentration in a lab scale cylindrical stirred PBR. The optimal growth for was found under CO_2 concentration of the range of 2.5–10 % (v/v). The developed model was verified with experimental data and can reproduce the measured trends of algal bioproductivity. With the model, the algal growth rate can be optimized through manipulation of the gas in-feed CO_2 concentration as well as the CO_2 outflow by reactor minimized. Also, the effect of mixing on dissolved CO_2 distribution can be evaluated revealing regions of excess and shortage, thus, the regions of decreased algae growth rate. Furthermore, bubble size can be optimized to support algae bioproductivity through efficient mixing and dissolution process. The developed CFD model can therefore serve as a valuable tool for evaluating PBR performance for CO_2 capture. Instead of being required to building new reactor geometries in order to evaluate their performance in real life experiments, the modelling approach allows to assess new designs in terms of e.g. regions with shortage of light or dissolved CO_2, thus providing a low-cost method for process and reactor design and optimization for CO_2 fixation.

CRediT authorship contribution statement

Ulla Ojaniemi: Writing – original draft, Visualization, Software, Methodology. **Anu Tamminen:** Investigation. **Jouni Syrjänen:** Software, Writing – review & editing. **Dorothee Barth:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.131715.

Data availability

Data will be made available on request.

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